1D Ring Two-Walker Encounter With Shortcut

Robust A1/A8 Verification and Bimodal Realization

valley-k-small report extension

March 9, 2026

Abstract

This report delivers the requested 1D ring encounter extension in two blocks: (1) an implementation-ready and numerically robust verification of Eq. (A1)⇔Eq. (A8) from the appendix "Product of two generating functions"; (2) an exact two-walker encounter solver on a 1D ring with a directed shortcut, including a shortcut-channel decomposition that explains the observed double-peak regime. All figures, tables, and machine-readable outputs are generated from one script.

1 Problem Setup

We study two independent lazy biased walkers on the same periodic 1D ring. Encounter first-passage is defined as

$$T_{\text{enc}} = \min\{t \geq 1 : X_t = Y_t\}, \qquad f_{\text{enc}}(t) = \mathbb{P}(T_{\text{enc}} = t).$$

Walker A has a directed shortcut src → dst with strength β; walker B has no shortcut.

Default representative parameters:

$$N = 101,\ q = 0.70,\ g_1 = 0.70,\ g_2 = -0.40,\ n_0 = 5,\ m_0 = 55,\ \text{src} = 5,\ \text{dst} = 70.$$

Directed Shortcut Injection (Selfloop Mode)

Walker A uses the same shortcut convention as the ring shortcut reports:

$$p_{\text{sc}} = \beta(1-q), \qquad 0 \leq p_{\text{sc}} \leq 1-q.$$

At node src we transfer this mass from self-loop to the directed edge:

$$P_1(\text{src}, \text{src}) = (1-q) - p_{\text{sc}},$$

$$P_1(\text{src}, \text{dst}) = P_1^{(0)}(\text{src}, \text{dst}) + p_{\text{sc}},$$

while all other entries follow the baseline lazy-biased kernel $P_1^{(0)}$. This keeps each row stochastic and preserves the interpretation of β as shortcut strength.

Unified Notation and Metrics

To make this first dedicated encounter report unambiguous, we fix all quantities used below:

- One-step kernels: P_1, P_2; non-absorbed joint law $J_t(i,j) = \Pr(X_t = i, Y_t = j, T_{\text{enc}} > t)$.
- Two encounter times:

$$T_{\text{any}} = \min\{t \geq 1 : X_t = Y_t\}, \quad T_\delta = \min\{t \geq 1 : X_t = Y_t = \delta\}.$$

- First-passage pmf and survival:

$$f(t) = \Pr(T = t), \quad S(t) = \Pr(T > t) = 1 - \sum_{\tau=1}^{t} f(\tau).$$

- Detector traces used for peak/valley metrics:

$$\bar{f}(t) = \mathrm{MA}_w[f](t),$$

with fixed odd window $w = 11$ for anywhere-encounter scans; for fixed-site K=2 diagnostics, use

$$\bar{\tilde{f}}(m) = \mathrm{MA}_9[\tilde{f}](m).$$

- Peak/valley detector times (under one fixed smoothing+threshold protocol):

anywhere: t_1 = first prominent local peak in $[t_{\text{ignore}}, t_{\text{end}}]$, $\quad t_2$ = selected comparison peak in $(t_1, t_{\text{end}}]$, $\quad t_v$

fixed-site K=2: run the same selector on parity index m of $g_m = \bar{\tilde{f}}(m)$, with $\tau \geq t_{\text{ignore,pair}}$, $\tau - t_1 \geq$ min_se

- Peak/valley intensity ratios (written on the detector trace g: anywhere uses $g = \bar{f}$, fixed-site K=2 uses $g = \bar{\tilde{f}}$ with the $t = 2m$ map):

$$R_{\text{peak}}(\tau) = \frac{\min\{g_{t_1}, g_\tau\}}{\max\{g_{t_1}, g_\tau\}}, \quad R_{\text{peak}} = R_{\text{peak}}(t_2), \qquad R_{\text{dir}} = \frac{g_{t_2}}{g_{t_1}}, \qquad R_{\text{valley}} = \frac{g_{t_v}}{\max\{g_{t_1}, g_{t_2}\}}.$$

For phase/onset and multi-peak score selection, we use R_{peak} (min/max) together with R_{valley}; R_{dir} is reported only as a directional diagnostic. By definition $R_{\text{dir}} = g_{t_2}/g_{t_1}$ may exceed 1, and this directional ratio is not used in phase thresholds. Candidate scoring uses $R_{\text{peak}}(\tau), R_{\text{valley}}(\tau)$, then commits to $R_{\text{peak}}, R_{\text{valley}}$ at $\tau = t_2$.

- Phase labels: single / weak / clear (encoded as 0/1/2); onset β means the first shortcut strength entering clear phase.

- Shortcut-channel shares:

$$h_{\text{sc}}(t) = \frac{f_{\text{sc}}(t)}{f_{\text{sc}}(t) + f_{\text{no}}(t)}, \quad H_{\text{sc}}(T) = \frac{\sum_{t \leq T} f_{\text{sc}}(t)}{\sum_{t \leq T} (f_{\text{sc}}(t) + f_{\text{no}}(t))}.$$

- Mass-conservation audit:

$$\varepsilon_{\text{mass}} = \left| 1 - \sum_{t=1}^{t_{\max}} f(t) - S(t_{\max}) \right|.$$

Mathematical Phase Definition (0/1/2)

Let g denote the detector trace ($g = \bar{f}$ for anywhere; $g = \bar{\tilde{f}}$ for fixed-site K=2 coarse diagnostics). Define prominent local-peak candidates

$$\mathcal{P} = \{\tau : \ g_{\tau-1} < g_\tau, \ g_\tau \geq g_{\tau+1}, \ g_\tau \geq 0.01 \max g\},$$

plus detector-window constraints:

$\tau \in [t_{\text{ignore}}, t_{\text{end}}]$ (anywhere), $\qquad \tau \geq t_{\text{ignore,pair}}, \ \tau - t_1 \geq$ min_sep_pair (fixed-site K=2 coarse).

If $|\mathcal{P}| < 2$, no valid peak pair is available and we set 'has_two=False'. We keep only candidates with $\tau - t_1 \geq 2$ so $t_v(\tau) = \arg\min_{t_1<t<\tau} g_t$ is defined on a strict interior interval (t_1, τ). Otherwise, we choose (t_1, t_2) by the configured selector:

$$t_1 = \min \mathcal{P}, \qquad t_2 = \arg \max_{\tau \in \mathcal{P},\ \tau > t_1} \frac{(\tau - t_1)\, R_{\text{peak}}(\tau)}{R_{\text{valley}}(\tau) + 10^{-12}},$$

where the fixed-site K=2 branch applies the additional parity-index constraint $\tau - t_1 \geq \text{min_sep_pair}$ and reports times after the $t = 2m$ back-map to the raw clock. If scores tie within numerical tolerance ($|\Delta\text{score}| \leq 10^{-12}$), we break ties by larger τ, then smaller R_{valley}, then larger R_{peak}. with

$$t_v(\tau) = \arg \min_{t_1<t<\tau} g_t, \quad R_{\text{peak}}(\tau) = \frac{\min\{g_{t_1}, g_\tau\}}{\max\{g_{t_1}, g_\tau\}}, \quad R_{\text{valley}}(\tau) = \frac{g_{t_v(\tau)}}{\max\{g_{t_1}, g_\tau\}}.$$

After selecting t_2, define

$$R_{\text{peak}} = R_{\text{peak}}(t_2), \qquad R_{\text{valley}} = R_{\text{valley}}(t_2).$$

The phase code is

$$\text{phase} = \begin{cases} 0, & \neg\text{has_two}, \\ 2, & \text{has_two} \ \wedge\ R_{\text{peak}} \geq \eta_p \ \wedge\ R_{\text{valley}} \leq \eta_v, \\ 1, & \text{has_two} \ \wedge\ \neg(R_{\text{peak}} \geq \eta_p \ \wedge\ R_{\text{valley}} \leq \eta_v)\,. \end{cases}$$

Parameter values used in this report:

$$(\eta_p, \eta_v) = (0.20, 0.90) \text{ for anywhere}, \qquad (\eta_p, \eta_v) = (0.10, 0.90) \text{ for fixed-site K=2 coarse}.$$

Anywhere primary detector settings are $(w, t_{\text{ignore}}, t_{\text{end}}) = (11, 80, 400)$. Fixed-site coarse settings are $(w, t_{\text{ignore,pair}}, \text{min_sep_pair}) = (9, 18, 8)$ in parity-coarsened index.

2 Compact Derivation Roadmap: A1 to A8

Define

$$\alpha_k = 1 + s_k = 1 + q\left(\cos\frac{2\pi k}{N} - 1\right), \qquad \widetilde{P}_{n_0}(n, u) = \sum_{k=0}^{N-1} \frac{h_k(n, n_0)}{1 - u\alpha_k}.$$

Insert into A1:

$$I(z) = \frac{1}{2\pi i}\oint_C \frac{du}{u}\widetilde{P}_{n_0}(n, u)\widetilde{P}_{m_0}\left(n, \frac{z}{u}\right) = \frac{1}{2\pi i}\oint_C \sum_{k_1,k_2} \frac{h_{k_1}h_{k_2}}{(1 - u\alpha_{k_1})(u - z\alpha_{k_2})}du.$$

For each (k_1, k_2), poles are

$$u = \frac{1}{\alpha_{k_1}} \quad \text{(outer poles)}, \qquad u = z\alpha_{k_2} \quad \text{(inner poles)}.$$

Choose $|u| = r$ such that

$$\max_k |z\alpha_k| < r < \min_{k:\alpha_k \neq 0}\left|\frac{1}{\alpha_k}\right|.$$

Then only inner poles are enclosed, and residue summation gives

$$I(z) = \sum_{k_1,k_2} \frac{h_{k_1}(n, n_0)h_{k_2}(n, m_0)}{1 - z\alpha_{k_1}\alpha_{k_2}},$$

which is exactly Eq. (A8).

Encounter Mapping (fixed-site and anywhere)

The same A1→A8 block is the core kernel for both encounter definitions:

- fixed-site encounter at δ:
$$Q_\delta(t) = P_{n_0}(\delta, t) P_{m_0}(\delta, t),$$
so A1/A8 applies directly at $n = \delta$.

- anywhere encounter:
$$Q_{\text{any}}(t) = \sum_{n=0}^{N-1} P_{n_0}(n, t) P_{m_0}(n, t),$$
hence
$$\widetilde{Q}_{\text{any}}(z) = \sum_{n=0}^{N-1} \frac{1}{2\pi i} \oint_C \frac{du}{u} \widetilde{P}_{n_0}(n, u) \widetilde{P}_{m_0}\Big(n, \frac{z}{u}\Big).$$

Using ring Fourier orthogonality, the modal double sum collapses to a single sum:

$$\widetilde{Q}_{\text{any}}(z) = \sum_{k=0}^{N-1} \frac{\frac{1}{N} e^{i2\pi k(m_0 - n_0)/N}}{1 - z\, \lambda_k^{(1)} \lambda_{-k}^{(2)}},$$

where $\lambda_k^{(1)}, \lambda_k^{(2)}$ are one-step mode eigenvalues of walkers A/B. So A1→A8 indeed covers both fixed-site and non-fixed encounter branches.

3 Numerically Stable Verification Routine

Use $u = re^{i\theta}$ and $du/u = i\, d\theta$:

$$I(z) = \frac{1}{2\pi} \int_0^{2\pi} \widetilde{P}_{n_0}(n, re^{i\theta}) \widetilde{P}_{m_0}\Big(n, \frac{z}{re^{i\theta}}\Big)\, d\theta \approx \frac{1}{M} \sum_{j=0}^{M-1} F\Big(re^{2\pi i j/M}\Big).$$

Implementation choices used in code:

1. enforce contour feasibility with $r_{\text{in}} < r < r_{\text{out}}$;
2. choose $r = \sqrt{r_{\text{in}} r_{\text{out}}}$ by default;
3. adaptive doubling in M from 16 upward until $|I_{2M} - I_M| < \varepsilon(1 + |I_{2M}|)$;
4. cross-check contour LHS against direct RHS (A8).

Four sanity cases (real/complex z, edge $q = 0$, small $|z|$) all pass with relative errors around $10^{-16} \sim 10^{-18}$.

Table 1: A1/A8 numerical verification summary.

Case	(N, q)	z	M	r	$\lvert LHS - RHS\rvert$	rel.err
R1_real_small	(16,0.30)	$0.40 + 0.00i$	128	0.632	8.13e-18	8.13e-18
R2_complex	(12,0.35)	$0.30 + 0.10i$	64	0.562	1.52e-17	1.52e-17
R3_edge_q0	(20,0.00)	$0.55 + 0.00i$	256	0.742	4.44e-16	1.38e-16
R4_small_z	(18,0.40)	$0.05 + 0.20i$	64	0.454	4.53e-18	4.53e-18

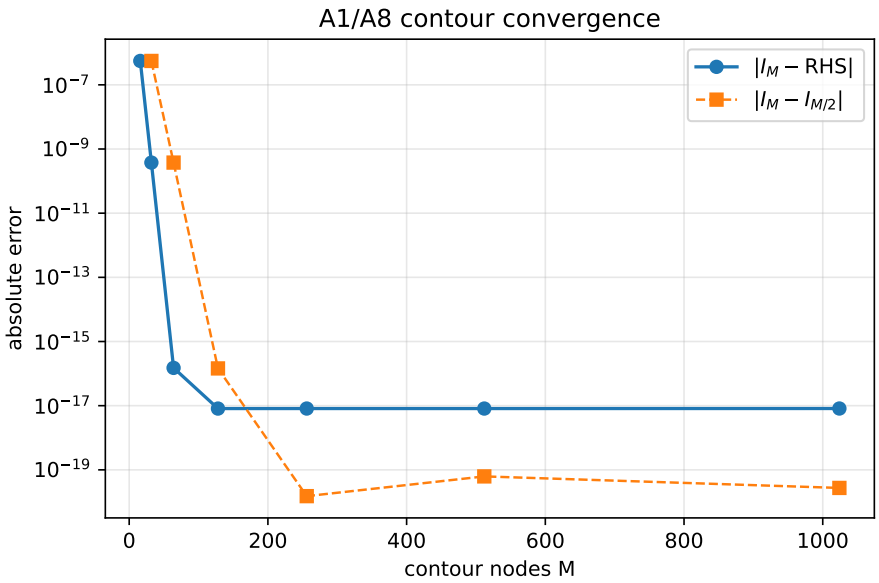

Figure 1: Error decay versus contour node count M.

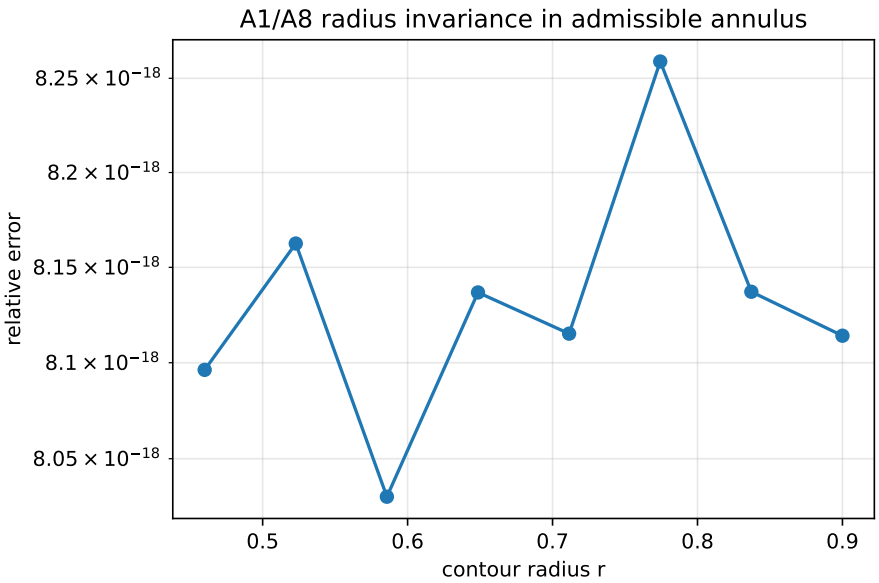

Figure 2: Radius invariance inside admissible annulus.

4 1D Ring Encounter Algorithm

Let

$$J_t(i,j) = \mathbb{P}(X_t = i, Y_t = j, T_{\mathrm{enc}} > t).$$

Exact update per time step:

$$J_{t+1} = P_1^\top J_t P_2, \qquad f_{\mathrm{enc}}(t+1) = \mathrm{tr}(J_{t+1}).$$

After extracting encounter flux, the diagonal of J_{t+1} is set to zero (absorbing condition). This gives deterministic, noise-free FPT curves.

A second coupled update tracks two channels:

- trajectories that have used the shortcut at least once;
- trajectories that have never used the shortcut.

Their sum equals the total encounter FPT and explains peak origins.

Mathematical Mechanism of Bimodality

In this 1D ring encounter model, the double-peak regime is produced by superposition of two channel-specific time scales rather than numerical artifacts:

$$f_{\text{enc}}(t) = f_{\text{sc}}(t) + f_{\text{no}}(t).$$

1. **Early lobe:** shortcut-used trajectories dominate around the first peak (high $h_{\text{sc}}(t)$), corresponding to shorter effective paths.
2. **Late lobe:** no-shortcut trajectories take over in the mid/late window and generate the second peak scale.
3. **Ring-topology amplification:** relative displacement and winding on a periodic ring increase delay/spread of the late channel, enlarging $t_2 - t_1$.

For the representative case (Table 5), we observe $t_1 < t_v < t_2$ with threshold-qualified $R_{\text{peak}}, R_{\text{valley}}$; R_{dir} is reported only as a directional diagnostic (not used in phase thresholds). Together with the channel handoff in the shortcut-share curve, this closes the mechanism chain for a clear-bimodal diagnosis.

5 Fixed-Site Encounter Extension ($n = m = \delta$)

We also complete the fixed-site encounter branch highlighted in your notes:

$$T_\delta = \min\{t \geq 1 : X_t = Y_t = \delta\}.$$

So we track two encounter definitions in one framework:

- anywhere encounter: $T_{\text{any}} = \min\{t \geq 1 : X_t = Y_t\}$;
- fixed-site encounter: $T_\delta = \min\{t \geq 1 : X_t = Y_t = \delta\}$.

For fixed-site, define the free-process synchronous occupancy

$$Q_{n_0,m_0}(t) = \Pr(X_t = \delta, Y_t = \delta \mid X_0 = n_0, Y_0 = m_0) = P^{(1)}_{n_0}(\delta, t)\, P^{(2)}_{m_0}(\delta, t).$$

Its generating function is exactly the appendix product-of-generating-functions object, evaluable by A1 contour integration or A8 modal double sum. Then renewal gives

$$\widetilde{f}_\delta(z) = \frac{\widetilde{Q}_{n_0,m_0}(z)}{\widetilde{Q}_{\delta,\delta}(z)}.$$

Numerically, we implement exact time-domain absorption at a single point (δ, δ) and apply K=2 parity coarse graining:

$$\widetilde{f}(0) = f(0), \qquad \widetilde{f}(m) = f(2m-1) + f(2m), \quad m \geq 1.$$

For first-passage traces, $f(0) = 0$ so diagnostics effectively start from $m \geq 1$; reported detector times use the right-endpoint map $t = 2m$.

At baseline $N = 30, q = 0.7, n_0 = 5, m_0 = 12, \delta = 0$, drift cases show:

- $(g_1, g_2) = (0, 0)$ is unimodal;
- $(0.9, 0.9)$, $(0.7, 0.7)$, and $(0.9, -0.9)$ are clearly bimodal.

The full phase-map occupancy (single/weak/clear fractions over the full drift grid) is auto-summarized in Table 2.

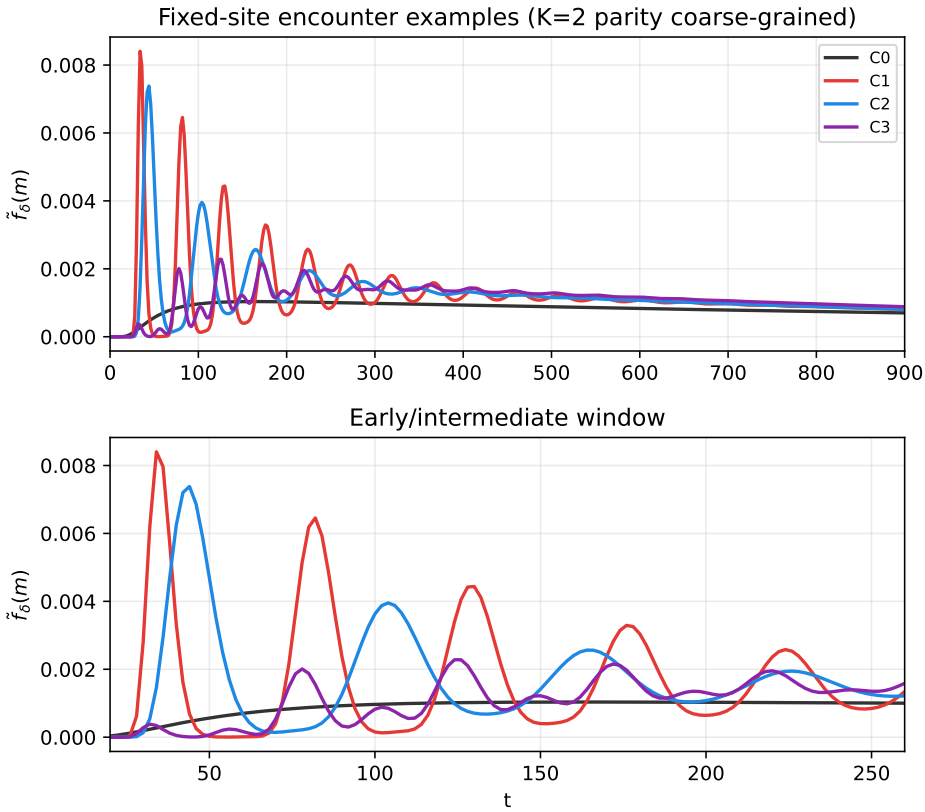

Figure 3: Fixed-site encounter examples after K=2 parity coarse graining.

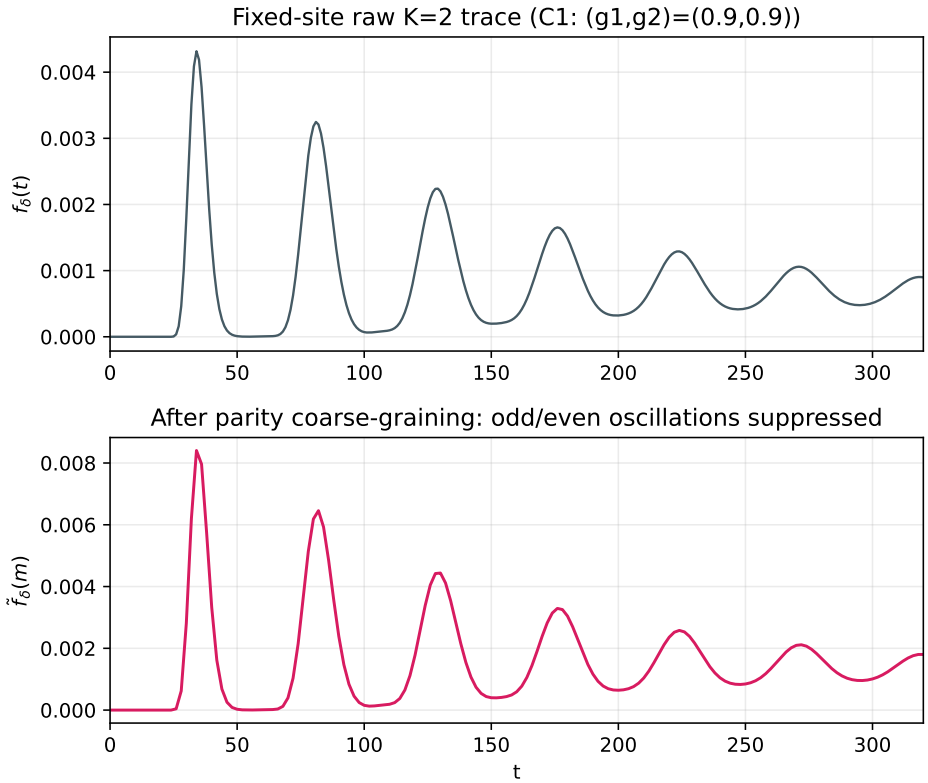

Figure 4: Why parity coarse graining is needed for K=2: raw trace (odd-even micro-oscillation) versus parity-coarsened trace used by the detector.

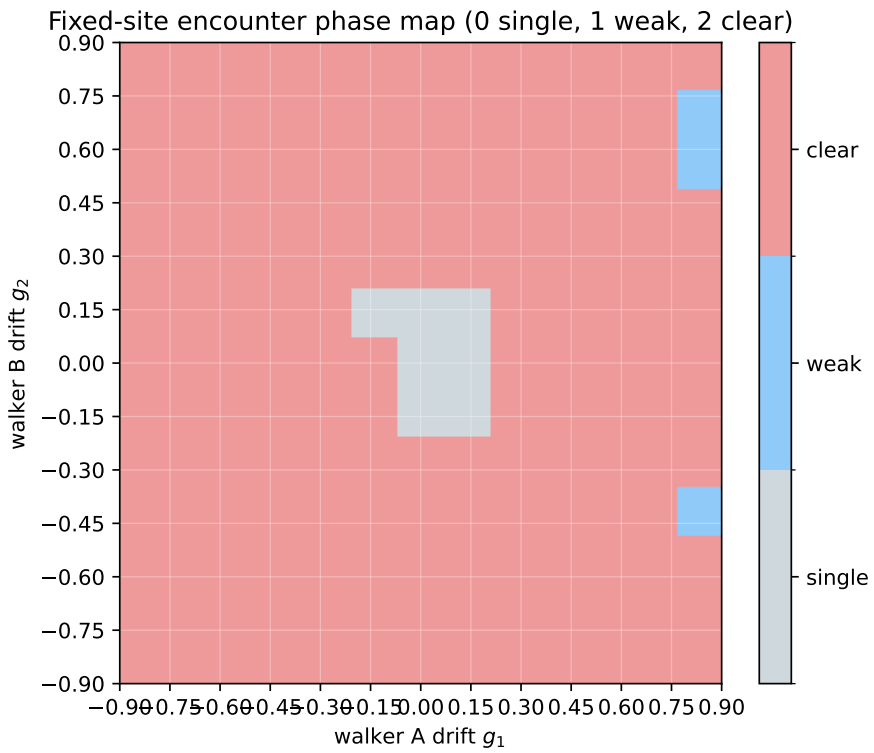

Figure 5: Fixed-site phase map on (g_1, g_2): 0 single, 1 weak, 2 clear.

Table 2: Fixed-site phase-map occupancy summary on the (g_1, g_2) grid.

Phase class	Count	Fraction
single (0)	7	0.041
weak (1)	3	0.018
clear (2)	159	0.941

The clear fraction is $159/169 = 0.941$ (auto-synced from generated scan data). This fixed-site benchmark is intentionally drift-dominated; K=2 parity coarse graining suppresses odd-even micro-structure so classification tracks separated time scales rather than parity artifacts. For first-passage traces, $f(0) = 0$, so $\widetilde{f}(0) = f(0) = 0$ and diagnostics effectively start at $m \geq 1$. As in the main text, candidate peak pairs require $\tau - t_1 \geq 2$ so $t_v(\tau) = \arg\min_{t_1<t<\tau} g_t$ is defined on a strict interior interval. This branch uses the same timescale selector as the main scan: detector trace $g_m = \widetilde{\bar{f}}(m)$, first peak + score-selected t_2 ($R_{\text{peak}}(\tau) = \frac{\min\{g_{t_1}, g_\tau\}}{\max\{g_{t_1}, g_\tau\}}$, $R_{\text{peak}} = \min / \max$ (i.e., $R_{\text{peak}}(t_2)$), $R_{\text{dir}} = g_{t_2}/g_{t_1}$ as a directional diagnostic only (it may exceed 1 and does not enter phase thresholds), $\text{score}(\tau) = \frac{(\tau - t_1)\, R_{\text{peak}}(\tau)}{R_{\text{valley}}(\tau) + 10^{-12}}$). If $|\Delta\text{score}| \leq 10^{-12}$, tie-break order is larger τ →smaller valley ratio→larger peak balance. With parity-index constraint $\tau - t_1 \geq \text{min_sep_pair}$, no extra t_{end} cutoff, and $t = 2m$ back-mapping.

Table 3: Fixed-site example metrics.

Case	(g_1, g_2)	phase	$t_1\,(= 2m)$	$t_2\,(= 2m)$	peak balance ratio ($\bar{\tilde{f}}$, min/max)	valley ratio ($\bar{\tilde{f}}$)
C0	(0.00,0.00)	0	-	-	–	–
C1	(0.90,0.90)	2	36	740	0.225	0.003
C2	(0.70,0.70)	2	44	518	0.210	0.035
C3	(0.90,-0.90)	2	78	592	0.864	0.424

In this table, t_1, t_2 are reported after mapping parity-coarsened indices back to the raw clock by $t = 2m$; both peak-balance and valley ratios are computed on the smoothed coarse trace $\bar{\tilde{f}}(m)$, so parity micro-oscillations do not enter fixed-site phase labels. When no second peak is detected (t_1, t_2 absent), both ratios are reported as "–" to mark the values as undefined rather than numeric 0/1.

6 Bimodality Results for 1D Ring Encounter

Scanning $\beta \in [0, 0.30]$ uses a fixed diagnostic protocol (same smoothing and $t_{\text{ignore}} = 80$). Detector-perturbation settings are fixed as smooth $\in \{9, 11, 13\}$, $t_{\text{ignore}} \in \{60, 80, 100\}$, min-ratio $\in \{0.18, 0.20, 0.22\}$, valley-cap $\in \{0.85, 0.90, 0.95\}$. Unless explicitly marked as a legacy comparison, onset/phase/representative metrics in this report use the timescale detector below as the primary selector.

Multi-Peak Selection Rule (implementation-level)

To avoid ambiguity when three or more prominent peaks coexist, we use the following deterministic rule: Let g denote the detector trace (anywhere: $g = \bar{f}$; fixed-site K=2: $g = \widetilde{\bar{f}}$ with $t = 2m$ mapping).

1. Anywhere scan: smooth the raw first-passage sequence into g with the fixed odd window, and keep candidate peaks in $[t_{\text{ignore}}, t_{\text{end}}]$.

2. Fixed-site K=2 coarse branch: run the same score-based selector on parity index m, with candidates constrained by $\tau \geq t_{\text{ignore,pair}}$ and $\tau - t_1 \geq$ min_sep_pair; this branch does not apply an extra t_{end} cutoff, and reported times are mapped back by $t = 2m$.

3. Set t_1 as the earliest prominent peak passing amplitude threshold.

4. For each later candidate $\tau > t_1$, define (candidate pairs are restricted to $\tau - t_1 \geq 2$ to keep the valley definition interior)

$$R_{\text{peak}}(\tau) = \frac{\min\{g_{t_1}, g_\tau\}}{\max\{g_{t_1}, g_\tau\}}, \quad t_v(\tau) = \arg\min_{t_1 < t < \tau} g_t, \quad R_{\text{valley}}(\tau) = \frac{g_{t_v(\tau)}}{\max\{g_{t_1}, g_\tau\}},$$

$$\text{score}(\tau) = \frac{(\tau - t_1) \cdot R_{\text{peak}}(\tau)}{R_{\text{valley}}(\tau) + 10^{-12}}.$$

5. Choose $t_2 = \arg\max_{\tau > t_1} \text{score}(\tau)$; if scores tie up to numerical tolerance ($|\Delta\text{score}| \leq 10^{-12}$), choose the larger τ (later timescale); if still tied, choose smaller valley ratio; if still tied, choose larger peak-balance ratio.

This "timescale detector" keeps the late-timescale peak stable in multi-peak windows and prevents non-physical t_2 switching.

Key numeric claims are inserted directly from generated outputs:

1. $\beta \leq 0.14$: phase 1 (weak/transition double structure);

2. coarse-onset from $\beta \approx 0.16$: phase 2 (clear bimodal);

3. representative case $\beta = 0.20$: peaks at $t_1 = 108$, $t_2 = 321$, peak-balance ratio $R_{\text{peak}} = 0.306$ (min/max), directed peak ratio $R_{\text{dir}} = 0.306$ (diagnostic only; may exceed 1; not used in phase thresholds), valley ratio $R_{\text{valley}} = 0.126$.

We further refine onset on $\beta \in [0.11, 0.21]$ with step 0.005, giving nominal clear-bimodal onset $\beta \approx 0.15$. Under detector perturbations, first-onset range is $[0.11, 0.16]$ with median 0.14; detector-agreement first crosses 50% at $\beta \approx 0.15$ and 75% at $\beta \approx --$, with agreement width $\Delta\beta_{25\to75} \approx --$. Mass-conservation check: $\sum_t f(t) = 0.99999994$, $S(t_{\text{max}}) \approx 5.88e-08$.

To keep iterative runs synchronized with narrative claims, key numeric indicators are auto-generated from machine outputs:

Table 4: Auto-generated key metrics for the 1D ring encounter scan.

Metric	Value
Coarse onset β	0.160
Refined onset β	0.150
Detector onset range	[0.110, 0.160]
Detector onset median	0.140
Agreement crossing (25%)	0.140
Agreement crossing (50%)	0.150
Agreement crossing (75%)	–
Agreement width (75% − 25%)	–
Agreement width (75% − 50%)	–
Representative peaks (t_1, t_2)	(108, 321)
Peak balance ratio (min/max) / valley ratio	0.306 / 0.126
Directed peak ratio $R_{\text{dir}} = \bar{f}(t_2)/\bar{f}(t_1)$ (diagnostic only)	0.306
Shortcut share around t_1 (± 20)	0.734
Shortcut share around valley (± 20)	0.596
Shortcut share around t_2 (± 20)	0.071
First $\leq 50\%$ shortcut-share time	155
Cumulative shortcut share at $t_{\max}$	0.121
Shortcut shift mass	0.060
Beta=0 relative-chain max diff	0.000
Onset scaling slope (log-beta vs N)	-0.0372
Onset scaling R^2	0.997
$\sum_t f(t)$ at $t_{\max}$	0.99999994
$S(t_{\max})$	5.885e-08

The table now includes an explicit $\beta = 0$ cross-check against the reduced relative-coordinate chain; the max absolute discrepancy is at machine precision and validates the exact two-walker solver baseline.

To provide a concrete 1D ring shortcut instance in report form, we isolate the representative setup below: its parameter tuple, detector outputs, and explicit peak annotations are shown together. The first three prominent peaks (after smoothing) are listed for multi-peak visibility, while the bimodality decision still uses the detector pair (t_1, t_2).

Table 5: Representative 1D ring + shortcut instance (parameters and peak diagnostics).

Field	Value
N / q	101 / 0.70
(g_1, g_2)	(0.70, -0.40)
Starts (n_0, m_0)	(5, 55)
Shortcut (src $\rightarrow$ dst)	($5 \rightarrow 70$)
Representative β	0.20
Detector pair (t_1, t_2, t_v)	(108, 321, 278)
Prominent peaks (P_1, P_2, P_3)	(108, 192, 321)
Peak balance ratio (min/max) / valley ratio	0.306 / 0.126
Directed peak ratio $R_{\text{dir}} = \bar{f}(t_2)/\bar{f}(t_1)$ (diagnostic only)	0.306

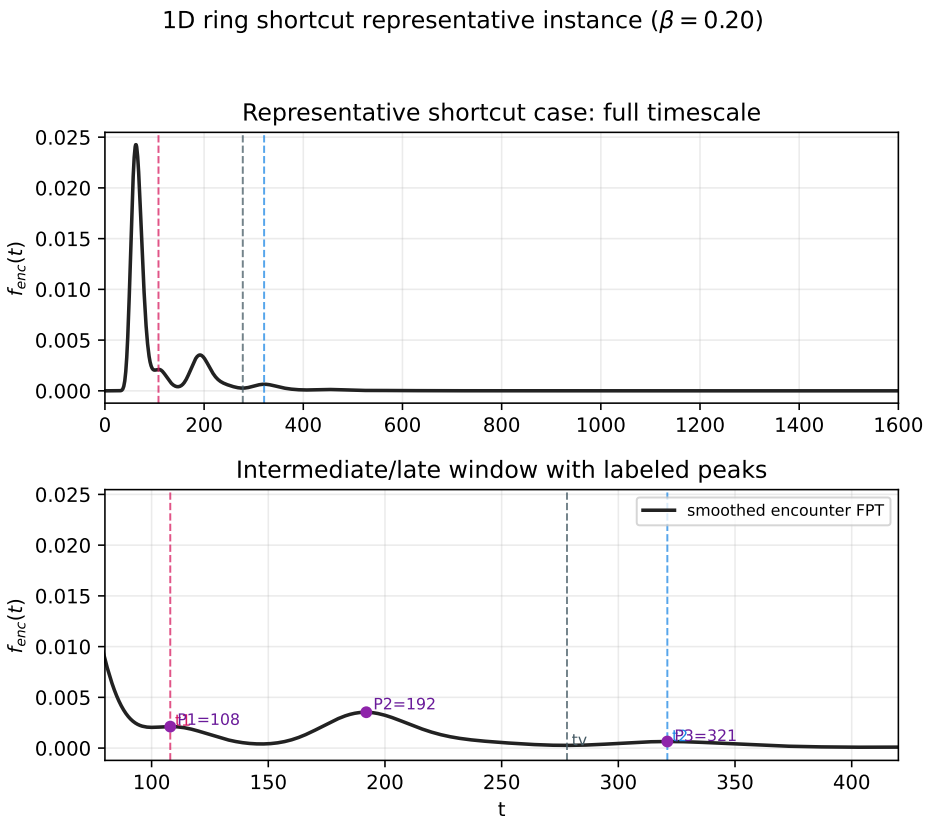

Figure 6: Representative case at $\beta = 0.20$ with labeled peak structure: detector pair (t_1, t_2) and prominent peaks P_1, P_2, P_3.

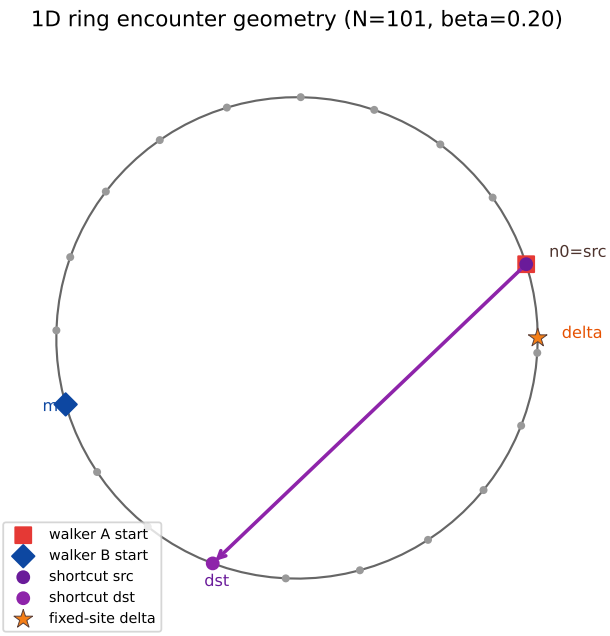

Figure 7: 1D ring geometry and shortcut placement; the fixed-site encounter point is marked as $\delta = 0$ (star).

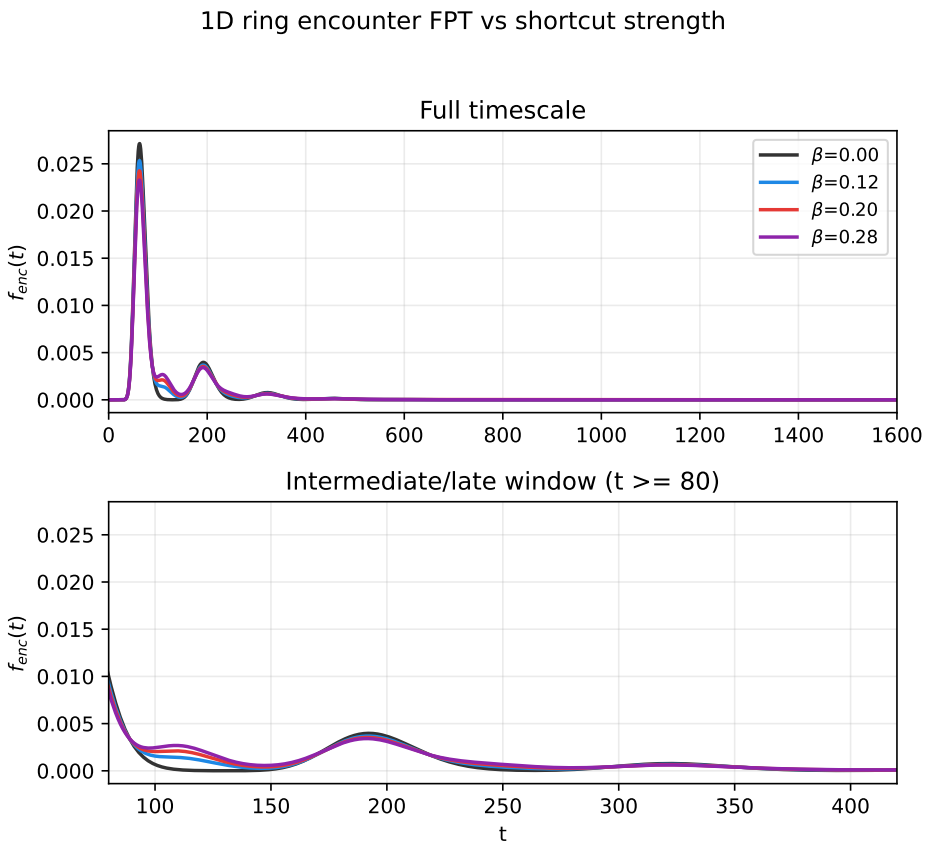

Figure 8: Encounter FPT curves across shortcut strengths.

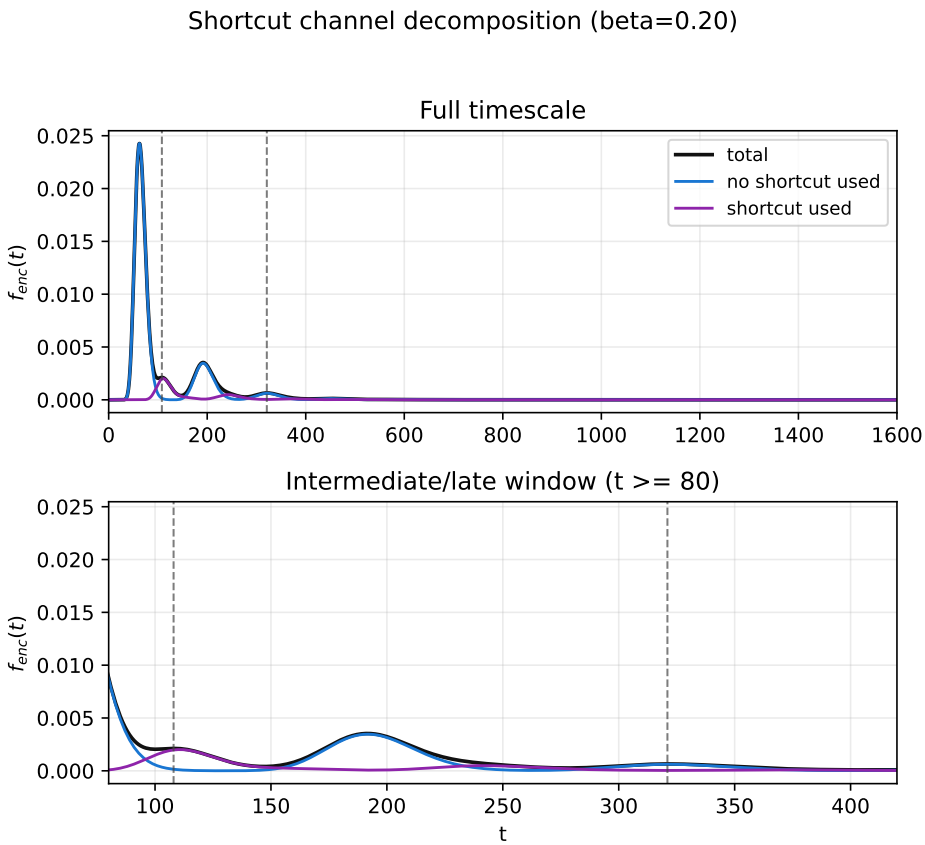

Figure 9: Shortcut-channel decomposition at $\beta = 0.20$: early lobe is shortcut-used dominated; late lobe is mostly no-shortcut trajectories.

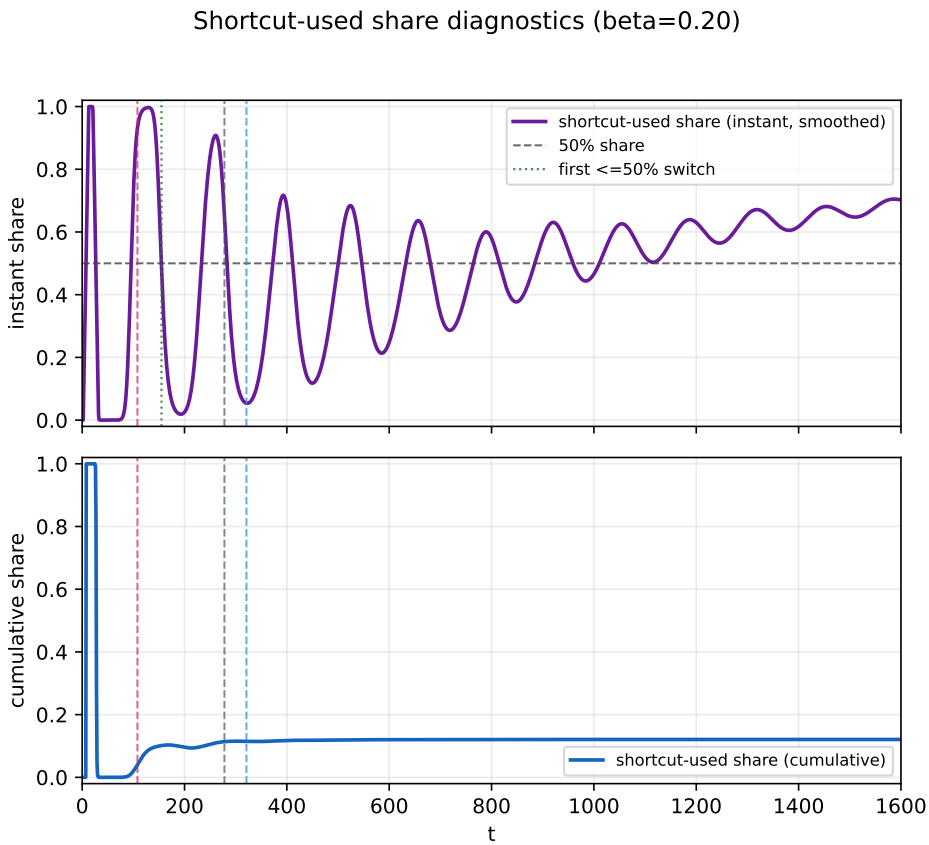

Figure 10: Shortcut-used share diagnostics at $\beta = 0.20$: instantaneous share (smoothed) and cumulative share versus time.

The share diagnostics make the channel handoff quantitative: shortcut-used trajectories dominate around the first peak and then give way to no-shortcut trajectories before the second peak, while the cumulative shortcut share remains bounded away from one.

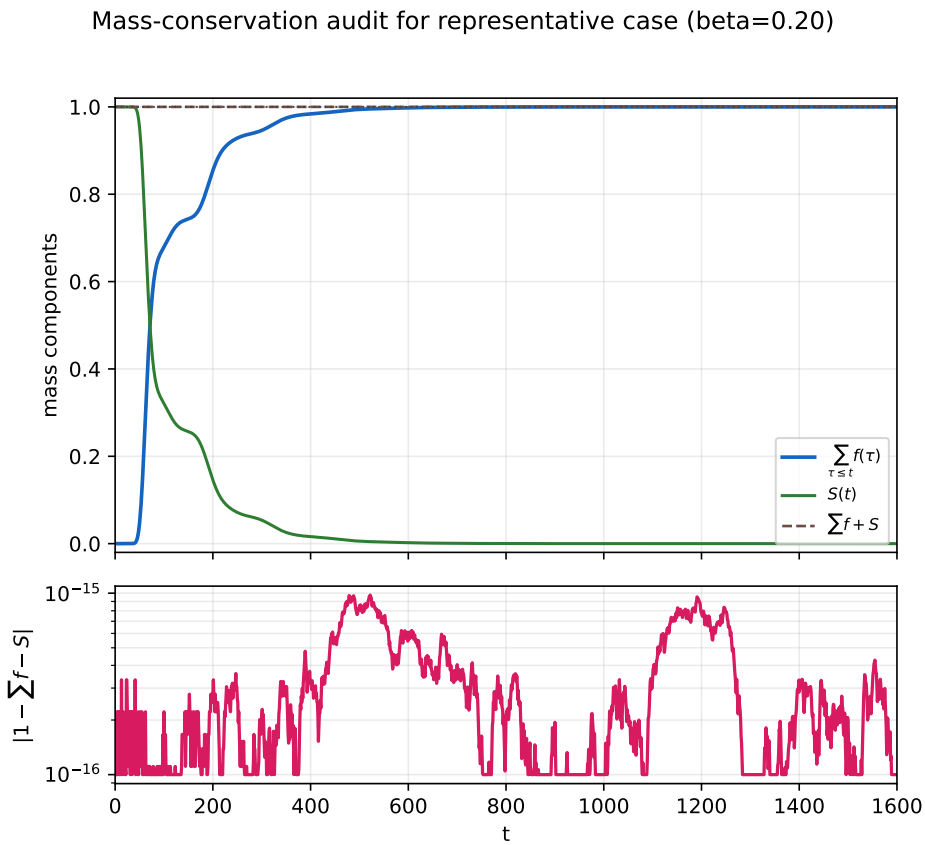

Figure 11: Mass-conservation audit for the representative case: cumulative encounter mass, survival tail, and residual $|1 - \sum_{\tau \le t} f(\tau) - S(t)|$.

The residual curve stays near machine precision and confirms that the late-time shoulder is a model effect rather than truncation leakage.

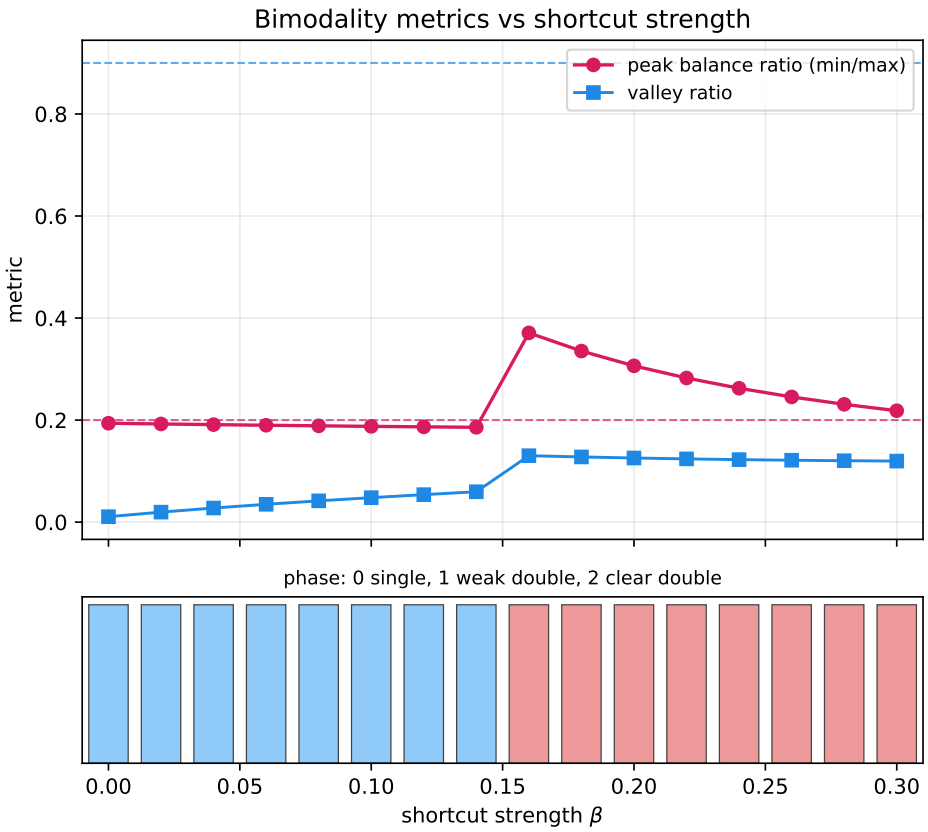

Figure 12: Peak-balance (min/max) / valley metrics and phase labels versus β (timescale detector, primary).

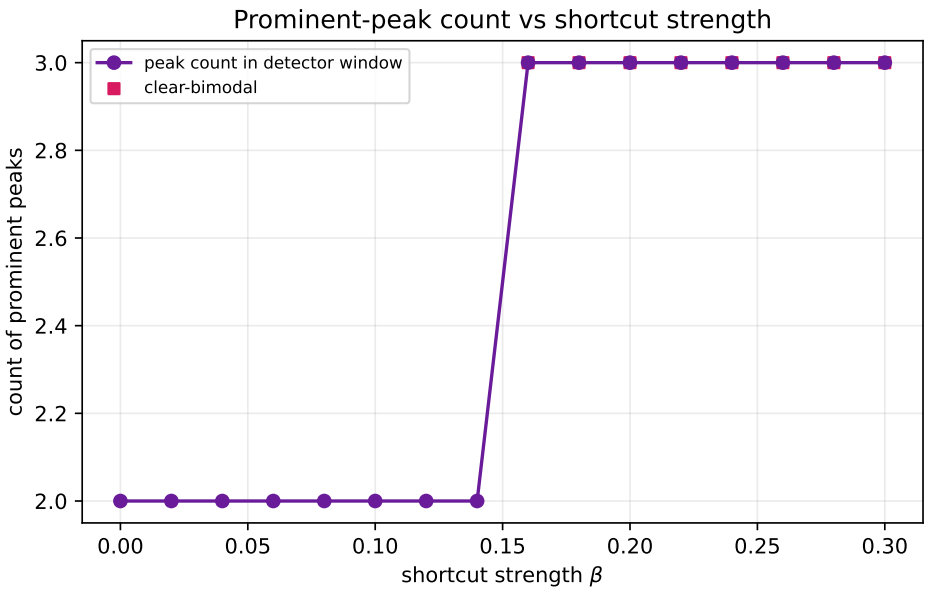

Figure 13: Number of prominent peaks in the detector window versus β (timescale detector view): the regime shifts from mostly 2 peaks to 3 peaks around onset.

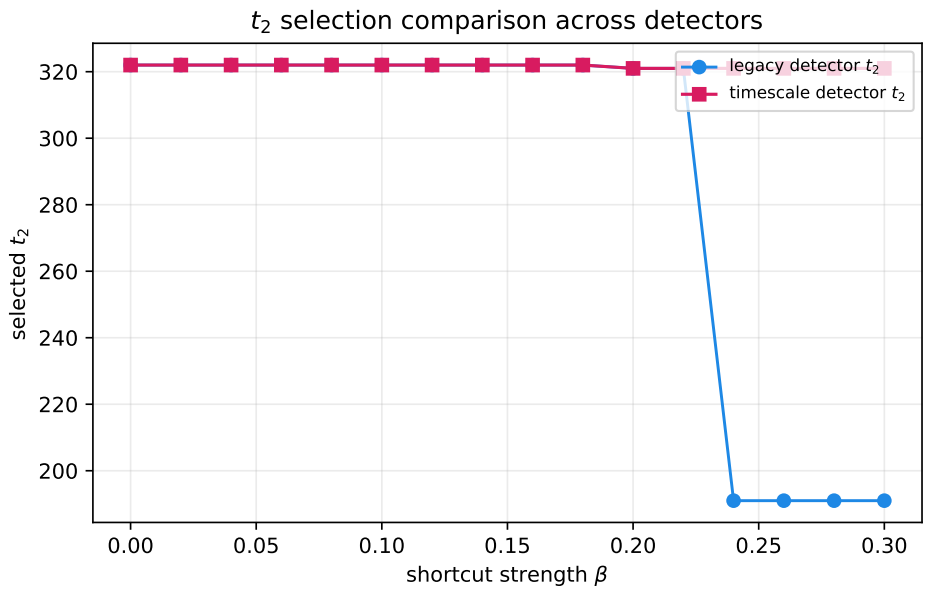

Figure 14: t_2 selection comparison: legacy second-peak rule can switch from late to middle peak near $\beta \approx 0.24$, while the timescale detector remains stable on the late scale.

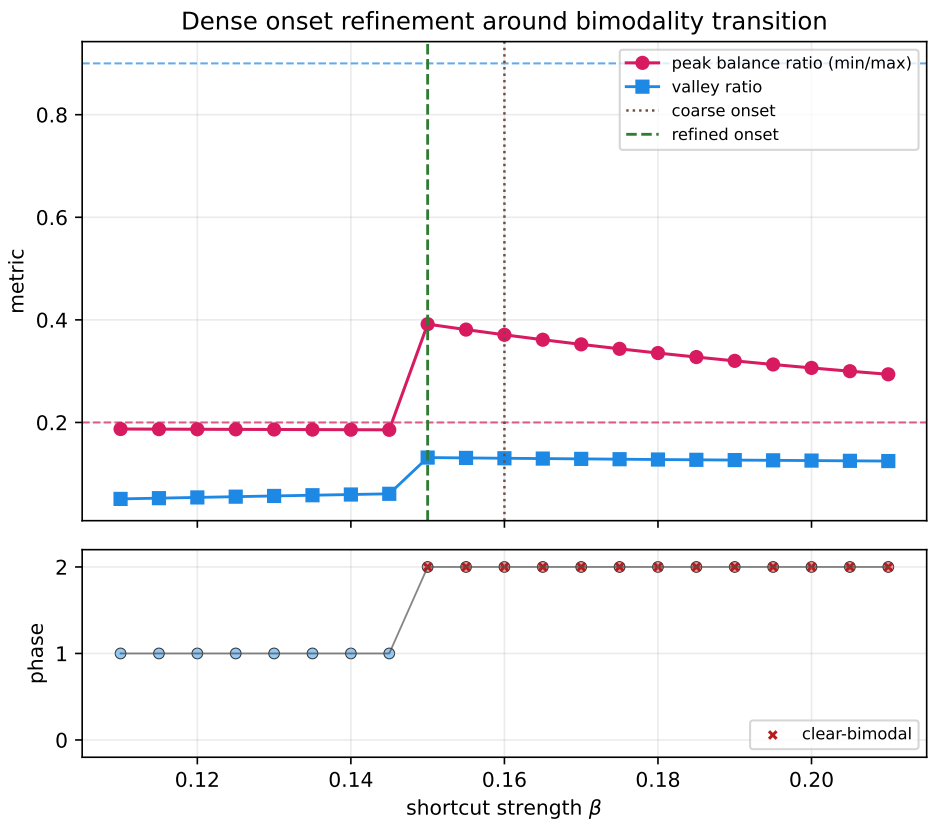

Figure 15: Dense refinement around onset: metric curves and phase labels on the local β window.

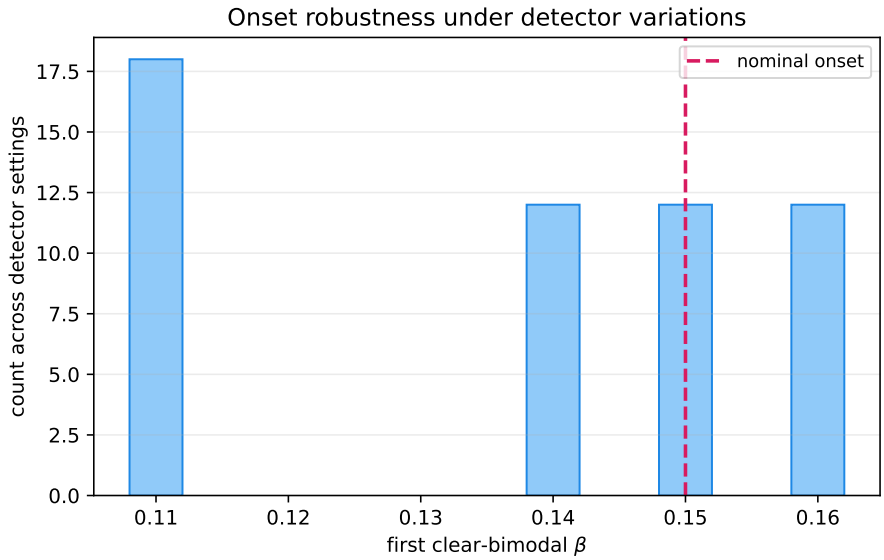

Figure 16: Distribution of first clear-bimodal onset under detector-parameter variations; dashed line is nominal refined onset.

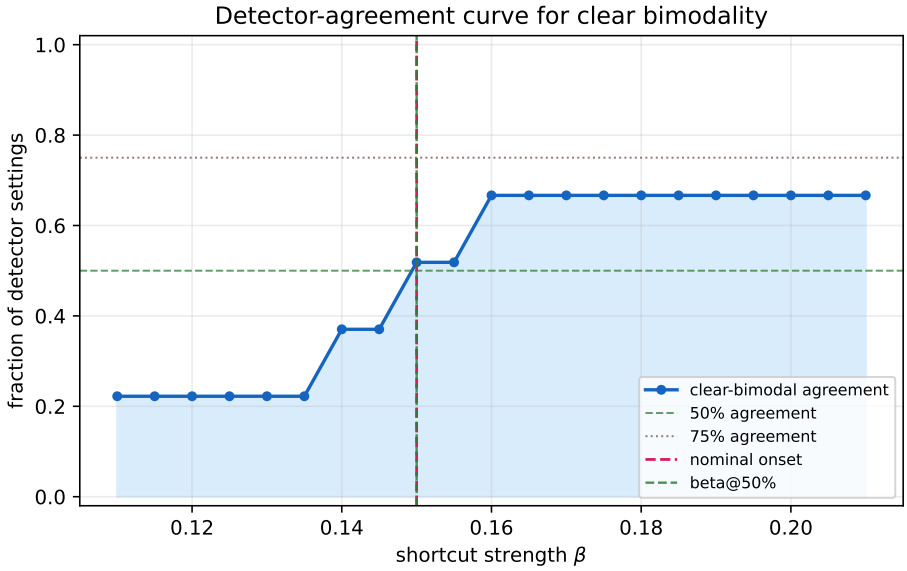

Figure 17: Detector-agreement curve across refined β: crossings at 25%, 50%, and 75% define onset location and transition width.

Ring-size robustness conclusions are also auto-generated to avoid manual drift: To avoid over-reliance on the single baseline size $N = 101$, we additionally scan $N \in \{81, 101, 121, 141\}$ with geometry-scaled $(n_0, m_0, \mathrm{src}, \mathrm{dst})$. For each N, we recompute coarse onset and clear-bimodal coverage on the nominal $\beta \in [0, 0.30]$ grid using the same timescale selector as the main scan (first peak + score-selected t_2), then compare representative peak/valley ratios at $\beta = 0.20$; if no clear onset appears, we extend once to $\beta \leq 0.50$ (step 0.02). Current run summary: 4 size points total, 3 with nominal-window onset, 1 recovered by extension, 0 still unresolved. Extension details: $N = 81$ is recovered only by extension at $\beta \approx 0.38$. Nominal-window conclusion: $N \in \{101, 121, 141\}$ all show clear onset inside the nominal window. The resulting trend is summarized in Fig. 18 and Table 6.

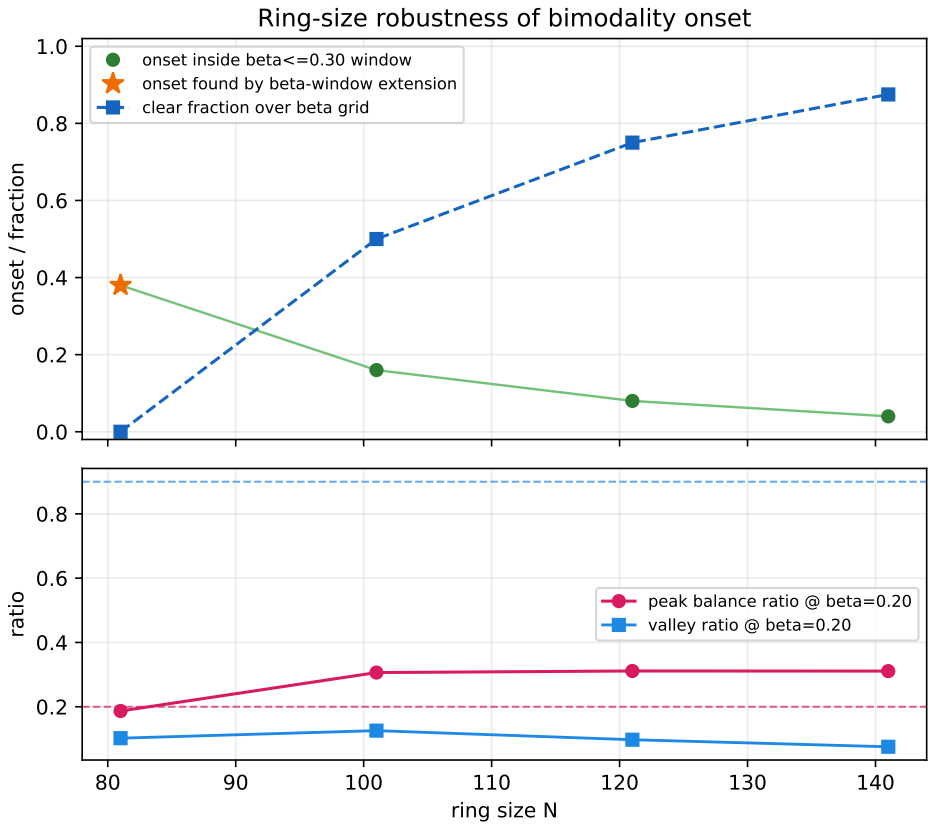

Figure 18: Ring-size robustness scan: onset and clear-fraction trends across N, plus representative ratio diagnostics at $\beta = 0.20$ (star marker = onset recovered by extension beyond $\beta = 0.30$).

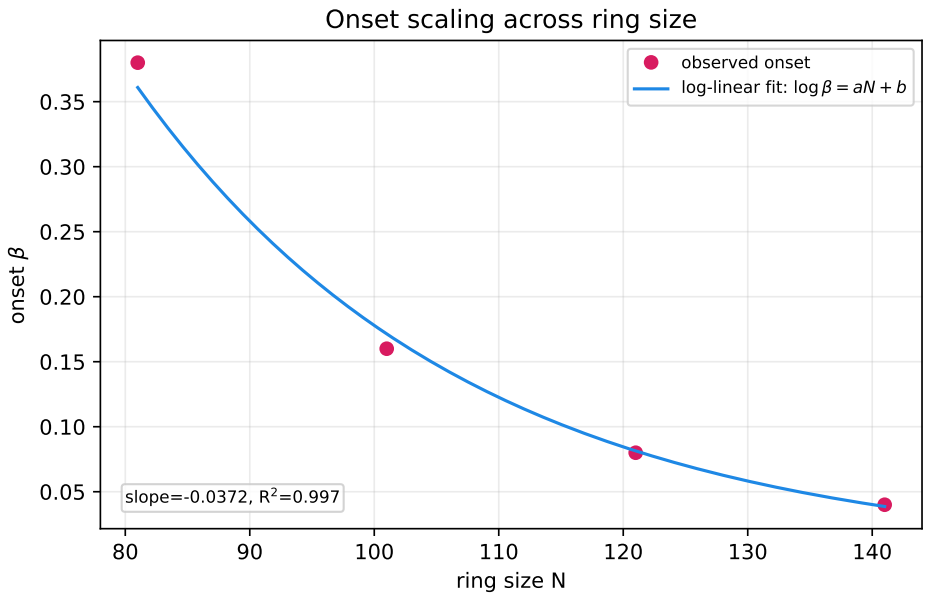

Figure 19: Onset scaling against ring size: linear fit in $(N, \log \beta_{\text{onset}})$ coordinates, with fit coefficients reported in Table 4.

To make extension usage explicit rather than implicit, Fig. 20 separates nominal-window detections from extension-recovered and unresolved cases, and overlays the actual per-N search

cap $\beta_{\max}(N)$. This avoids interpreting "no onset in $[0, 0.30]$" as a hard physical absence when it can be a window effect. Table 6 now also exposes the per-row onset source tag ('main/extended/none') and the corresponding $\beta_{\max}$, so extension recovery can be audited without cross-checking raw CSV files.

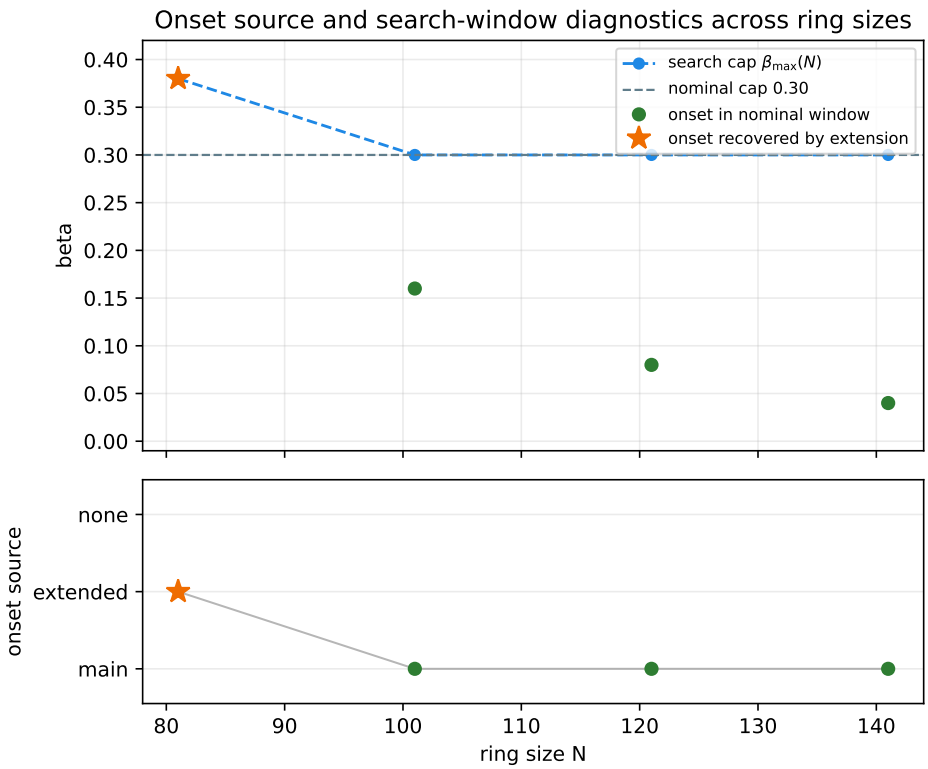

Figure 20: Onset-source diagnostics across ring sizes: top panel shows resolved onset versus per-N search cap and nominal cap 0.30; bottom panel labels whether each onset is found in the nominal window, recovered by extension, or still unresolved.

Table 6: Ring-size onset robustness summary with explicit onset source and search cap columns ($\beta^\dagger$ = resolved onset; "*" marks extended onset; "$> x$" means no clear onset up to search cap x).

N	onset $\beta^\dagger$	onset source	$\beta_{\max}$	clear frac	phase@0.20	peak@0.20	valley@0.20
81	0.380^*	`extended`	0.380	0.000	1	0.187	0.102
101	0.160	`main`	0.300	0.500	2	0.306	0.126
121	0.080	`main`	0.300	0.750	2	0.311	0.097
141	0.040	`main`	0.300	0.875	2	0.311	0.076

Table 7: Shortcut scan metrics and phase labels.

β	phase	t_1	t_2	peak balance ratio (min/max)	valley ratio	bimodal
0.00	1	192	322	0.194	0.011	0
0.02	1	192	322	0.192	0.020	0
0.04	1	192	322	0.191	0.028	0
0.06	1	192	322	0.190	0.035	0
0.08	1	192	322	0.189	0.042	0
0.10	1	192	322	0.188	0.048	0
0.12	1	192	322	0.187	0.054	0
0.14	1	192	322	0.186	0.060	0
0.16	2	107	322	0.371	0.130	1
0.18	2	108	322	0.335	0.128	1
0.20	2	108	321	0.306	0.126	1
0.22	2	109	321	0.283	0.124	1
0.24	2	109	321	0.262	0.122	1
0.26	2	109	321	0.245	0.121	1
0.28	2	109	321	0.231	0.120	1
0.30	2	109	321	0.218	0.120	1

7 Pitfalls and Debug Checklist

1. **Contour crossing poles:** abort if $r_{\text{in}} \geq r_{\text{out}}$.
2. **Near-singular amplification:** monitor $\min_{k_1,k_2} |1 - z\alpha_{k_1}\alpha_{k_2}|$.
3. **False early micro-peaks:** keep fixed smoothing and t_{ignore}.
4. **Truncation bias:** always report $\sum_t f(t)$ and survival tail.

8 Positioning Against Luca (Bristol) Literature

This report places encounter dynamics into the same finite-domain random-walk first-passage framework used in Luca Giuggioli's line of work:

- The fixed-site renewal ratio
$$\widetilde{f}_\delta(z) = \frac{\widetilde{Q}_{n_0,m_0}(z)}{\widetilde{Q}_{\delta,\delta}(z)}$$
is consistent with multi-target/heterogeneous first-passage decompositions[2].
- The A1/A8 generating-function product route is compatible with defect/heterogeneity-oriented analytical strategies in related first-passage studies[1].

So the encounter report is aligned with literature at the level of definitions, measurable diagnostics, and analytic-numeric closure, while adding a shortcut-channel evidence chain specific to this project.

9 Reproducibility

```
cd reports/ring_two_walker_encounter_shortcut
python3 code/two_walker_ring_encounter_report.py
```

```
latexmk -xelatex -interaction=nonstopmode -halt-on-error -auxdir=build \
  -emulate-aux-dir ring_two_walker_encounter_shortcut_cn.tex
latexmk -pdf -interaction=nonstopmode -halt-on-error -auxdir=build \
  -emulate-aux-dir ring_two_walker_encounter_shortcut_en.tex
```

References

[1] S. Sarvaharman and L. Giuggioli, *First-passage-time statistics of a random walker under biased quenched disorder*, Phys. Rev. E **102**, 062124 (2020). DOI: 10.1103/PhysRevE.102.062124.

[2] L. Giuggioli, S. Sarvaharman, D. Das, D. Marris, and T. Kay, *Multi-target search in bounded and heterogeneous environments: a lattice random walk perspective*, arXiv:2311.00464v2 [cond-mat.stat-mech] (2024).